



Web Fundamentals

WEF1VO

Teil 2: HTML von A bis Z

Medientechnik und -design | WS 2025/26

Last Time on **WEF1**



```
<!DOCTYPE html>
<html lang="de">
  <head>
    <title>Titel der Seite</title>
  </head>
  <body>
    <p>Inhalt der Seite</p>
  </body>
</html>
```

Kopfdaten

Nicht sichtbare Inhalte einer Webseite

Alles im Kopf

```
  
↓  
https://www.johndoe.com/image.png
```



```
<head>
```

```
<title>John Doe's Blog</title>
```

```
<base href="https://www.johndoe.com/">
```

```
<meta name="description" content="John Doe's Blog">
```

```
<meta http-equiv="refresh"  
      content="20;URL=page2.html">
```

```
<meta http-equiv="content-type"  
      content="text/html; charset=utf-8">
```

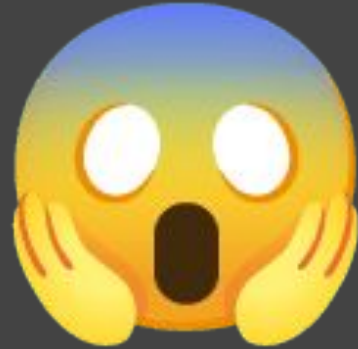
```
<meta charset="utf-8">
```

```
</head>
```



UTF-8

**8-Bit
UCS (Universal-Coded Character
Set)
Transformation
Format**



Struktur in HTML

Gliederung von Dokumenten und Texten

Header



`<header>`

`<hgroup>`

`<h1>John Doe's Blog</h1>`

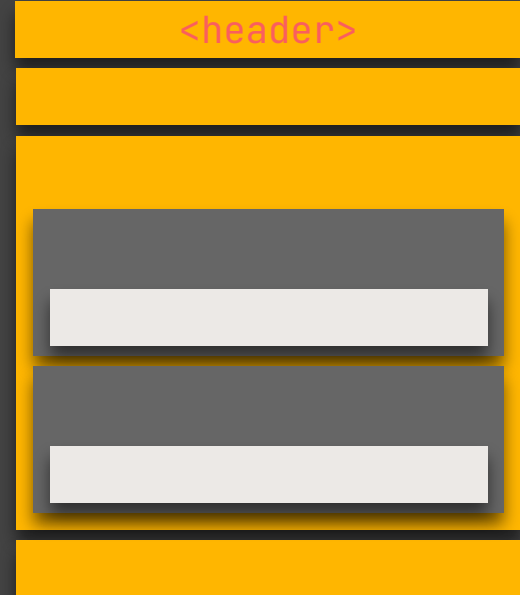
`<p>Gedanken eines Namenlosen</p>`

`</hgroup>`

`</header>`



`<header>`



Footer

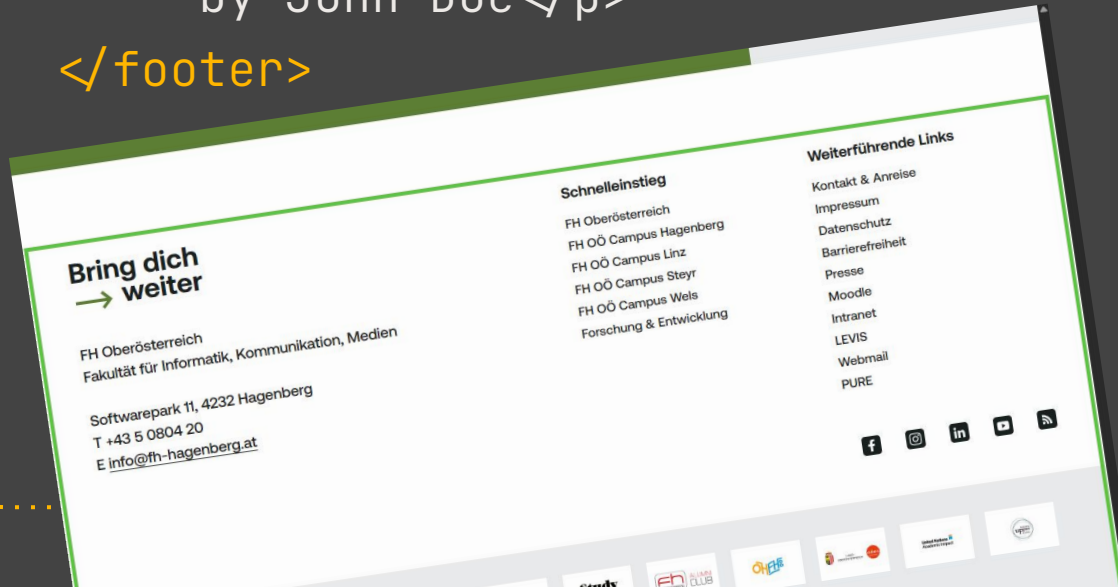


`<footer>`

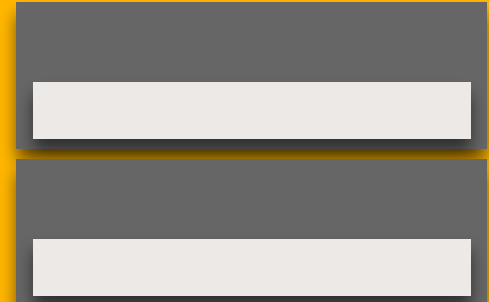
`<p>© 2025`

`by John Doe</p>`

`</footer>`



`<header>`



`<footer>`

Navigation



`<nav>`

`<ul>`

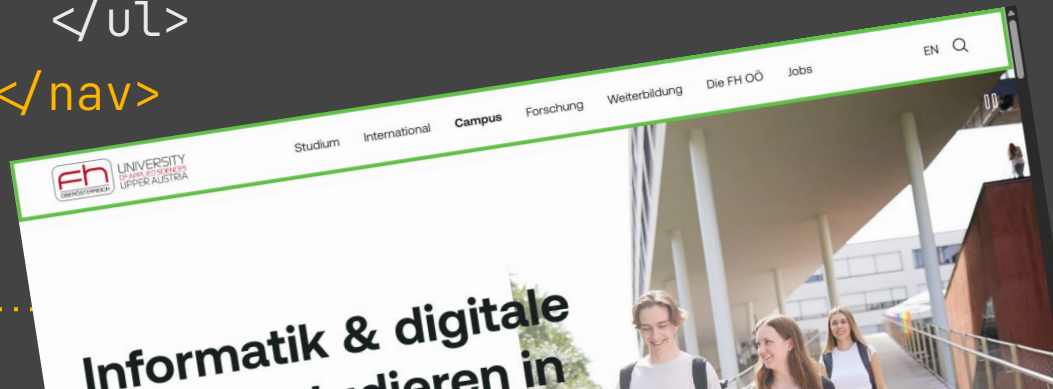
`<li><a href="...">Home</a></li>`

`<li>...</li>`

`<li>...</li>`

`</ul>`

`</nav>`



`<header>`

`<nav>`



`<footer>`

Artikel



```
<article>
```

```
  <hgroup>
```

```
    <h2>Die Hagenberg Story</h2>
```

```
    <p>Alles über den Studienort</p>
```

```
  </hgroup>
```

```
  <p>Es begann alles 1993 ...</p>
```

```
</article>
```



```
<header>
```

```
<nav>
```

```
<article>
```

```
<article>
```

```
<footer>
```

Abschnitte

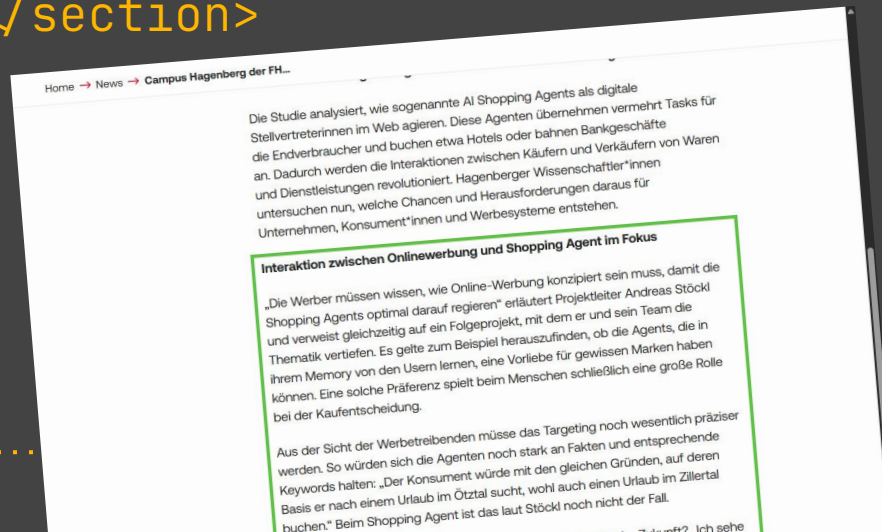


<section>

<h2>Aktuelles</h1>

<p>Seit kurzem ... </p>

</section>



<header>

<nav>

<article>

<section>

<article>

<section>

<footer>

Hauptbereich

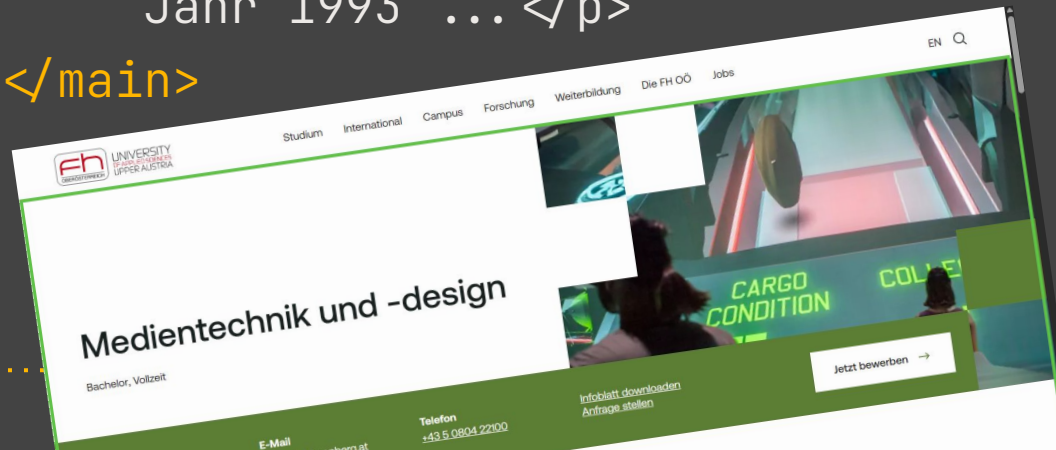


`<main>`

`<h1>Die Geschichte von
Hagenberg</h1>`

`<p>Es begann alles im
Jahr 1993 ...</p>`

`</main>`



`<header>`

`<nav>`

`<main>`

`<article>`

`<section>`

`<article>`

`<section>`

`<footer>`

Asides & Adressen



<aside>

<p>Hagenberg ist
eine Gemeinde...</p>

</aside>

<address>

This blog is run by

<a href="mailto:
john.doe@johndoe.com">

John Doe

</address>



**Bring dich
→ weiter**

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Fakultät für Informatik, Kommunikation, Medien

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T +43 5 0804 20
E info@fh-hagenberg.at

Überschriften & Outline



```
<h1>Überschrift A</h1>
```

```
<h2>Überschrift B</h2>
```

```
<h3>Überschrift C</h3>
```

```
<h2>Überschrift D</h2>
```

```
<h2>Überschrift E</h2>
```

```
<h1>Überschrift F</h1>
```

1. Überschrift A
 - 1.1. Überschrift B
 - 1.1.1. Überschrift C
 - 1.2. Überschrift D
 - 1.3. Überschrift E
2. Überschrift F

```
<h1>John Doe's Blog</h1>
```

```
<section>
```

```
<h2>Artikel zu Katzen</h2>
```

```
<article>
```

```
<h3>Eine Katze namens Nobody</h3>
```

```
<p>Es war einmal...</p>
```

```
</article>
```

```
</section>
```

1. John Doe's Blog
 - 1.1. Artikel zu Katzen
 - 1.1.1. Eine Katze namens Nobody

Überschriftengruppen



```
<article>
  <header>
    <hgroup>
      <h2>Die Hagenberg Story</h2>
      <p>Alles über den Studienort im Mühlviertel</p>
    </hgroup>
  </header>
  <p>Es begann alles im Jahr 1993 ...</p>
</article>
```

Suchbereich



```
<search>
```

```
  <form action="..." method="get">
    <label for="suche">Suche:</label>
    <input type="search" id="suche">
    <button type="submit">Suchen</button>
  </form>
```

```
</search>
```


Liste mal drei



```
<ul>
```

```
  <li>1. Eintrag</li>
```

```
  <li>2. Eintrag</li>
```

```
  <li>3. Eintrag</li>
```

```
</ul>
```

```
<ol>
```

```
  <li>1. Eintrag</li>
```

```
  <li>2. Eintrag</li>
```

```
  <li>3. Eintrag</li>
```

```
</ol>
```

```
<dl>
```

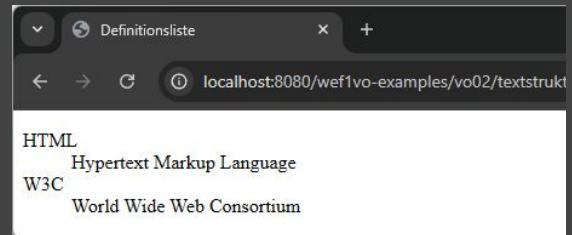
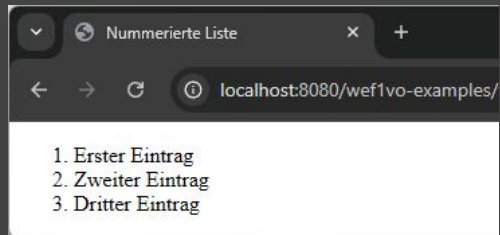
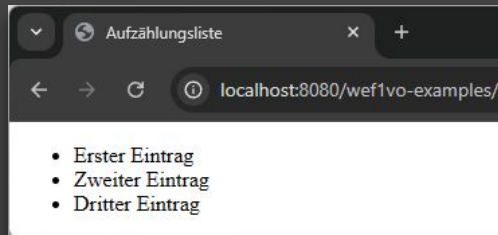
```
  <dt>HTML</dt>
```

```
  <dd>Hypertext Markup  
      Language</dd>
```

```
  <dt>W3C</dt>
```

```
  <dd>World Wide Web  
      Consortium</dd>
```

```
</dl>
```



Mehr Gruppierung



`<p>`

Absatz

`<div>`

Allgemeiner Bereich

`<pre>`

Vorformatierter Bereich

`<hr>`

Horizontale Linie

`<blockquote cite="...">`

Zitat mit Quelle

Links und Elemente auf der Textebene

Struktur innerhalb von Text



Verweise im Text

```
<a href="datei1.html">Ein Link</a>
```

```
<a href="docs/datei2.html">Ein Link</a>
```

```
<a href=" ../ ../datei3.html">Ein Link</a>
```

Relativ

```
<a href="https://www.unicode.org/">
```

```
Ein HTTPS-Link</a>
```

Absolut



Links mit Slash

Slash: Ausgehend vom Root-Verzeichnis. Wird zu root-relativem Link!

<https://johndoe.com/blog/index.html>

Ohne führenden Slash

```
<a href="media/projekte.html">  
  Projekte</a>
```



[https://johndoe.com/
blog/media/projekte.html](https://johndoe.com/blog/media/projekte.html)

Mit führendem Slash

```
<a href="/media/projekte.html">  
  Projekte</a>
```



[https://johndoe.com/
media/projekte.html](https://johndoe.com/media/projekte.html)



Mehr Links

```
<a href="file:///C:/Dokumente/HTML/index.html">  
  Ein File-Link</a>
```

File

```
<h1 id="anfang">John Doe's Blog</h1>
```

```
<!-- <h1><a id="anfang">John Doe's Blog</a></h1> -->
```

...

```
<p><a href="#anfang">Zum Anfang </a></p>
```

Intern



E-Mail -Adressen

```
<a href="mailto:john.doe@johndoe.com">  
  Send an e-mail!</a>
```

Einfach

```
<a href="mailto:john.doe@johndoe.com?  
  cc=survey@johndoe.com&  
  subject=Feedback%20to%20your%20site&  
  body=Dear%20John... ">  
  Send Feedback!</a>
```

Erweitert

Elemente auf der Textebene



`<em>`

Betonung

`<strong>`

Wichtigkeit

`<small>`

Kleingedrucktes

`<s>`

Ungültiges

`<cite>`

Titel von Werken

`<q>`

Zitatsteil

`<dfn>`

Definition

`<abbr>`

Abkürzung

`<data>`

Maschinendaten

`<time>`

Datum/Uhrzeit

`<code>`

Quellcode

`<var>`

Math. Variablen

`<samp>`

Programmoutput

`<sub>/<sup>`

Hoch-/Tiefstellung

`<mark>`

Hervorhebung

`<span>`

Generischer Bereich

Bilder

Putting Media into Hypermedia

Bilder & Abbildungen



```

```

```

```



```
<figure>  
    
  
  <figcaption>  
    Das HTML5-Logo.  
  </figcaption>  
</figure>
```



Das HTML5-Logo.



Respect Pixel Density

```

```

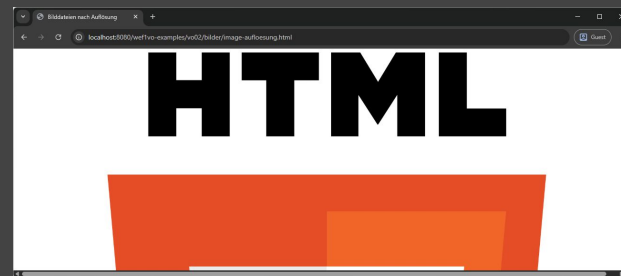
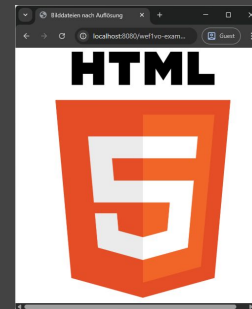




What about Resolutions ?

```

```





And Proper

Sizes?

```

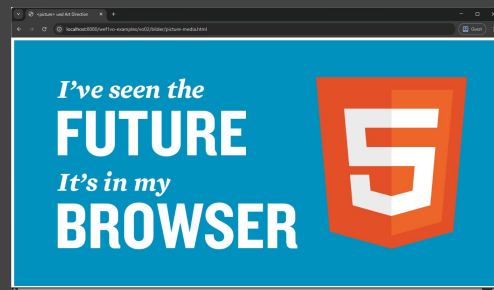
```





Art Direction

```
<picture>  
  <source media="(min-width: 990px)" srcset="html5sticker.png 1800w">  
    
</picture>
```





Raster- grafik im Web

GIF

256 Farben
Verlustfrei
Transparenzen

G

PNG

16,7 Millionen Farben
Verlustfrei
Transparenzen

P

16,7 Millionen Farben
Verlustbehaftet
Ideal für Fotos

JPEG

J

16,7 Millionen Farben
Verlustfrei & -behaftet
Bessere Kompr. als JPEG

WEBP

W

AVIF

16,7 Millionen Farben
Verlustfrei & verlustbehaftet
Beste Kompression von allen, aber etwas langsamer

A



Vektorgrafik im Web

XML-basiertes Format

Einbindung über `<img>`-Tag

Inline möglich (SVG-Markup direkt in der HTML-Datei).



Formulare

Daten in Webseiten übermitteln



Gut in **<form>**

```
<form action="mailto:john.doe@johndoe.com"  
      method="post" enctype="text/plain"  
      name="feedback" autocomplete="no" novalidate>
```

...

```
</form>
```

Formulare beschriften n



```
<label for="some-name">Name:</label>  
<input type="text" id="some-name">
```



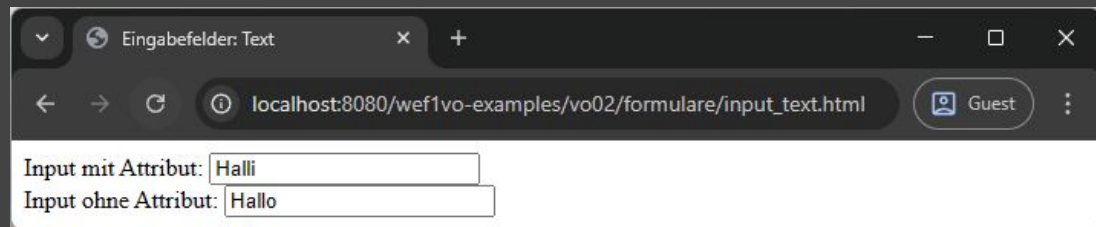
Einzeiliger Input

```
<input type="..."  
      size="..."  
      value="..."  
      name="..."  
      maxLength="..."  
      readonly>
```

Text



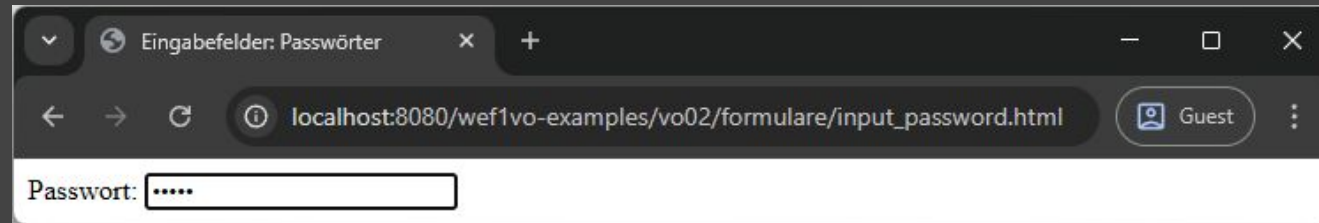
```
<label for="with">Input mit Attribut:</label>  
<input type="text" id="with"><br>  
<label for="without">Input ohne Attribut:</label>  
<input id="without">
```



Passwörter



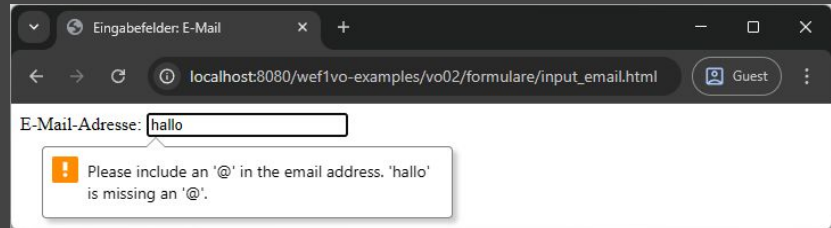
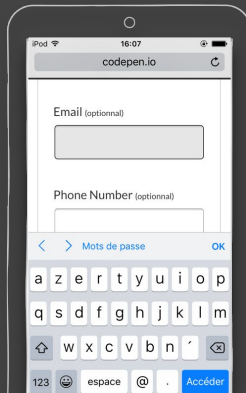
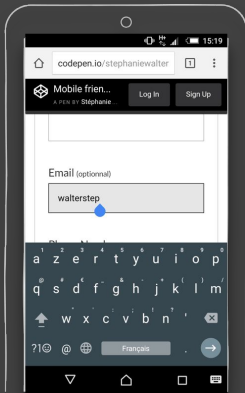
```
<label for="pass">Passwort:</label>  
<input type="password" id="pass">
```



E-Mail-Adressen



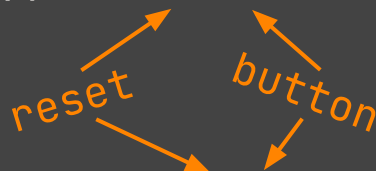
```
<label for="e-mail">E-Mail-Adresse:</label>  
<input type="email" id="e-mail">
```



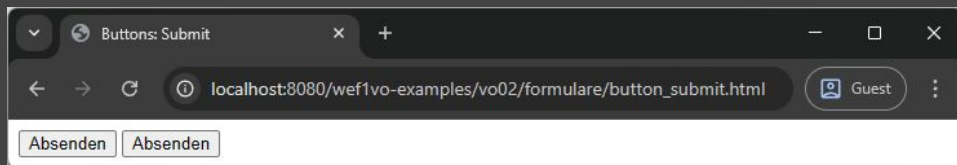


Buttons

```
<input type="submit" value="Absenden">
```



```
<button type="submit">Absenden</button>
```





Mehrzeiliger

`<label for="comment">Kommentar:</label>`

`<textarea name="text" rows="5" id="comment"></textarea>`

`<button type="submit">Senden</button>`

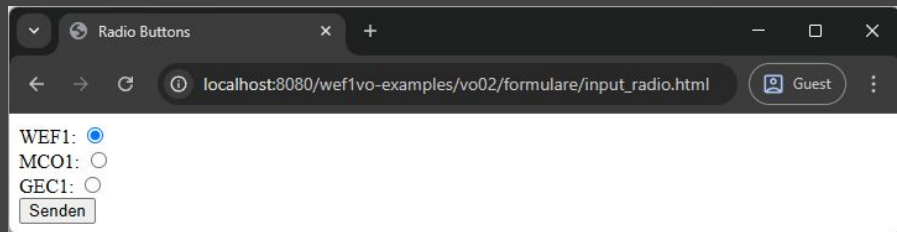
The screenshot shows a web browser window with the title 'Mehrzeiliges Eingabefeld'. The address bar shows 'localhost:8080/wef1vo-examples/vo02/formulare/textarea.html'. The form contains a text area with the text 'Hallo, wie geht es dir? Etwas längerer Text. Hier wurde Enter gedrückt.' and a 'Senden' button. The text area is labeled 'Kommentar:'.

text=Hallo, wie geht es dir? Etwas längerer Text.
Hier wurde Enter gedrückt.

Radio Buttons



```
<label for="wef">WEF1:</label>
<input type="radio" name="favorit" value="WEF1" id="wef" checked><br>
<label for="mco">MCO1:</label>
<input type="radio" name="favorit" value="MCO1" id="mco"><br>
<label for="gec">GEC1:</label>
<input type="radio" name="favorit" value="GEC1" id="gec"><br>
<button type="submit">Senden</button>
```



Checkboxen



```
<label for="wef">WEF1:</label>
<input type="checkbox" value="WEF1" id="wef" checked><br>
<label for="mco">MCO1:</label>
<input type="checkbox" value="MCO1" id="mco"><br>
<label for="gec">GEC1:</label>
<input type="checkbox" value="GEC1" id="gec"><br>
<button type="submit">Senden</button>
```

WEF1: ☒

MCO1: ☐

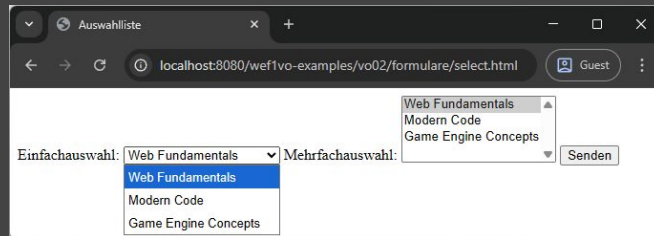
GEC1: ☐

Senden

Auswahllisten



```
<label for="mehrfach">Mehrfachauswahl:</label>
<select name="faecher" id="mehrfach" multiple>
  <option value="WEF1" selected>Web Fundamentals</option>
  <option value="MC01">Modern Code</option>
  <option value="GEC1">Game Engine Concepts</option>
</select>
<button type="submit">Senden</button>
```



Custom Auswahl



```
<label for="customized">Mehrfachauswahl:</label>
```

```
<select name="faecher3" id="customized">
```

```
  <button><selectedcontent></selectedcontent></button>
```

```
  <option value="WEF1" selected>Web Fundamentals
    <strong>WEF1</strong></option>
```

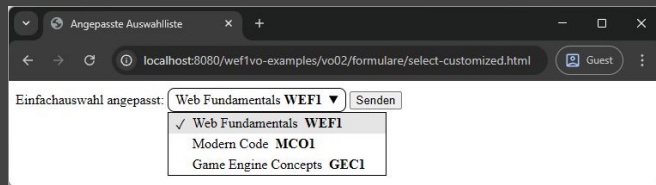
```
  <option value="MC01">Modern Code
    <strong>MC01</strong></option>
```

```
  <option value="GEC1">Game Engine Concepts
    <strong>GEC1</strong></option>
```

```
</select>
```

```
<button type="submit">Senden</button>
```

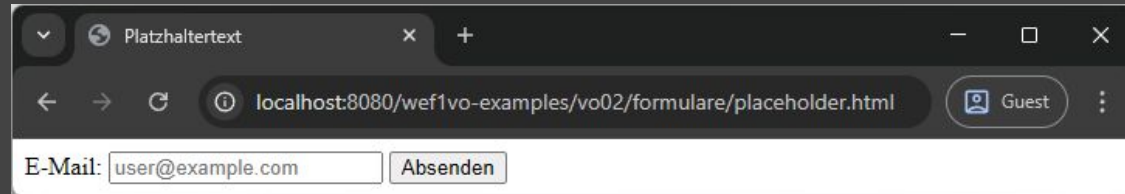
```
select,
::picker(select) {
  appearance: base-select;
}
```



Platzhaltertext



```
<label for="contact">E-Mail:</label>  
<input type="email" id="contact"  
      placeholder="user@example.com">
```



Autofokus & Pflichtfelder

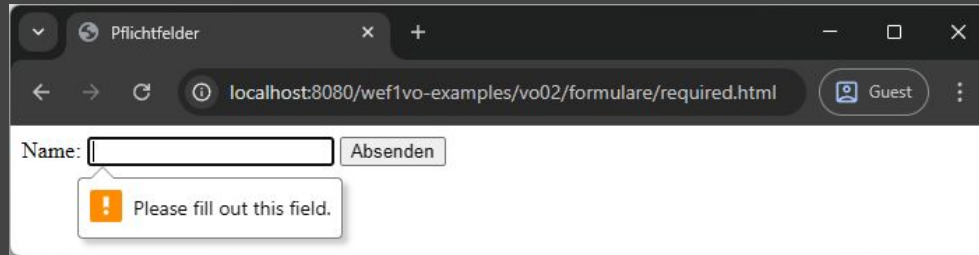


```
<label for="searchbox">Search:</label>
```

```
<input type="search" id="searchbox" autofocus>
```

```
<label for="name">Name:</label>
```

```
<input type="text" id="name" required>
```



Formulare strukturieren



```
<fieldset>
  <legend>Daten</legend>
  <label for="name">Name:</label>
  <input type="text" id="name"><br>
  <label for="contact">E-Mail:</label>
  <input type="email" id="contact">
</fieldset>
<button type="submit">Absenden</button>
```

Gruppierung mit Fieldsets

localhost:8080/wef1vo-examples/vo02/formulare/fieldset.html

Daten

Name:

E-Mail:

Absenden



Formulare – The Big Picture

The screenshot shows a web browser window with the address bar displaying `localhost:8080/wef1vo-examples/vo02/formulare/formular_komplett.html`. The page title is "Vollständiges HTML-Grundger...". The form is titled "Feedback" and contains the following elements:

- Meine Daten:** A fieldset containing two single-line text inputs: "Name: John Doe" and "E-Mail: user@example.com".
- Kommentar und Stimmung:** A fieldset containing a multi-line text area with the placeholder text "Super Beitrag!".
- Kommentare:** A label for the text area.
- Ich bin...:** A label for the radio buttons.
- Radiobuttons:** Three radio buttons labeled "happy", "neutral", and "traurig".
- Submit/Reset:** Two buttons at the bottom labeled "Kommentar posten" and "Zurücksetzen".

Annotations with orange arrows point to the following elements:

- Einzeiliger Input:** Points to the "Name" input field.
- Mehrzeiliger Input:** Points to the "Kommentar und Stimmung" text area.
- Submit/Reset:** Points to the "Kommentar posten" button.
- Fieldsets & Labels:** Points to the "Meine Daten" and "Kommentar und Stimmung" fieldsets.
- Platzhaltertext:** Points to the "Super Beitrag!" placeholder text.
- Radiobuttons:** Points to the "happy", "neutral", and "traurig" radio buttons.

Eine **HTML-Seite** mit Form



```
<!DOCTYPE html>
<html lang="de">
<head>
  <title>HTML-Grundgerüst mit Formular</title>
  <meta charset="utf-8">
</head>
<body>
<h1>Feedback</h1>
<!-- Formular -->
</body>
</html>
```

Eine HTML-Seite mit Form



```
<form action="mailto:john.doe@johndoe.com" method="post" enctype="text/plain">

  <fieldset>
    <legend>Meine Daten</legend>
    <label for="autor">Name:</label>
    <input type="text" name="autor" id="autor" size="20" placeholder="John Doe" required><br>
    <label for="email">E-Mail:</label>
    <input type="email" name="email" id="email" size="20" placeholder="user@example.com"><br>
  </fieldset>

  <fieldset>
    <legend>Kommentar und Stimmung</legend>
    <label for="kommentare">Kommentare:</label>
    <textarea name="kommentare" id="kommentare" rows="8" cols="20"
      placeholder="Super Beitrag!" required></textarea><br>
    <p>Ich bin...</p>
    <label for="happy">happy</label>
    <input type="radio" name="stimmung" value="happy" id="happy">
    <label for="neutral">neutral</label>
    <input type="radio" name="stimmung" value="neutral" id="neutral">
    <label for="traurig">traurig</label>
    <input type="radio" name="stimmung" value="traurig" id="traurig">
  </fieldset>

  <button type="submit">Kommentar posten</button> <button type="reset">Zurücksetzen</button>

</form>
```



Nächstes Mal in WEF1

Cascading Stylesheets (CSS)-Grundlagen

Credits



- S. 36: Phone Screenshots: <https://www.smashingmagazine.com/2018/08/ux-html5-mobile-form-part-2/>